Universidad de los Andes

Cálculo Integral con Ecuaciones (202319)

Ejercicios para practicar

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11.5 Series alternantes

(1) Determine si la serie converge o diverge.

(a) $\displaystyle\sum_{n=0}^{\infty}(-1)^n\frac{n}{2n^3+1}$

(b) $\displaystyle\sum_{n=0}^{\infty}(-1)^n\frac{4+3^n}{2^n}$

(c) $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{n^2-1}{3n^4+1}$

(d) $\displaystyle\sum_{n=0}^{\infty}\frac{(-1)^n}{1+2^n}$

(e) $\displaystyle\sum_{n=2}^{\infty}(-1)^n\frac{\sin^2 n}{n^3}$

(f) $\displaystyle\sum_{n=0}^{\infty}\frac{(-1)^n}{\sqrt{4n^4+1}}$

(g) $\displaystyle\sum_{n=3}^{\infty}(-1)^n\frac{\cos^2 n}{n^3+1}$

(h) $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{n^2}{n^3+4}$

(i) $\displaystyle\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}$

(j) $\displaystyle\sum_{n=1}^{\infty}(-1)^n\cos(\pi/n)$

(k) $\displaystyle\sum_{n=1}^{\infty}\left(-\frac{n}{5}\right)^n$

(l) $\displaystyle\sum_{n=4}^{\infty}(-1)^{n-1}\frac{\ln n}{n}$

(2) Encuentre los valores de p para los cuales la serie converge.

(a) $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^p}$

(b) $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n+p}$